

# **Unit 5 :: Fundamentals of Data Visualization**

❧ SEC-03-PHY-91-R-03 ❧

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# Syllabus

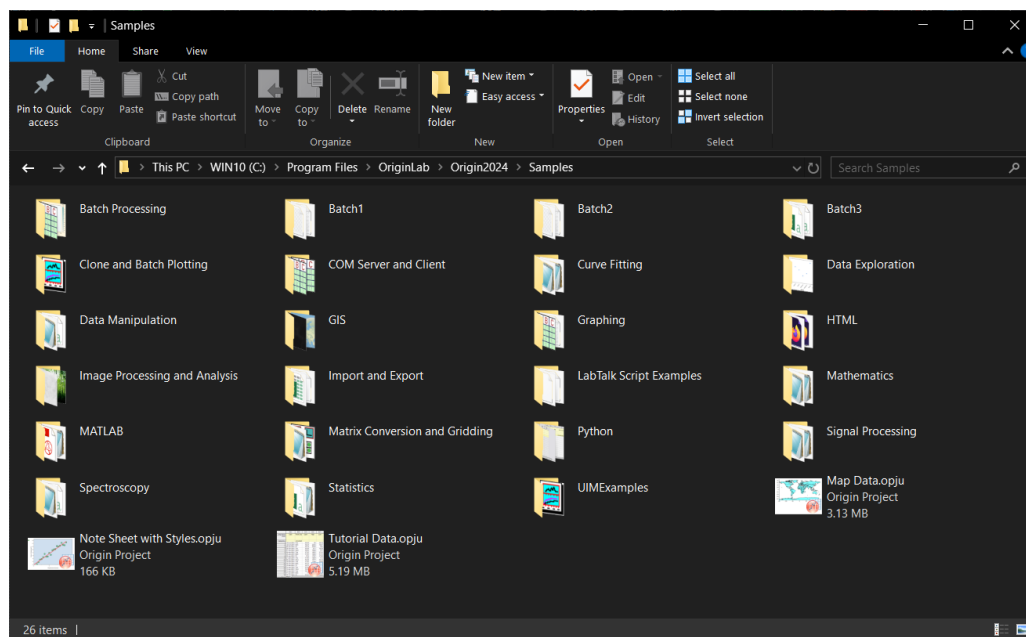
## Unit 5: Fundamentals of data visualization

Principles and use cases for categorical and continuous data plots, including column, line, pie, bar, area and scatter plots; customizing chart elements (axes, scales, legends, and labels); fine-tuning aesthetics for clarity and technical accuracy; advanced plotting techniques: 3D surface plots, contour mapping and multi-layered/multipanel figures.

# 1 Fundamentals of plotting in Origin

An installation of Origin comes with a collection of sample data which is recommended for a beginner to learn the fundamentals of plotting and subsequent customization. In order to access this collection of sample data, one can go to the main menu and click Help > Open Folder > Sample Folder.

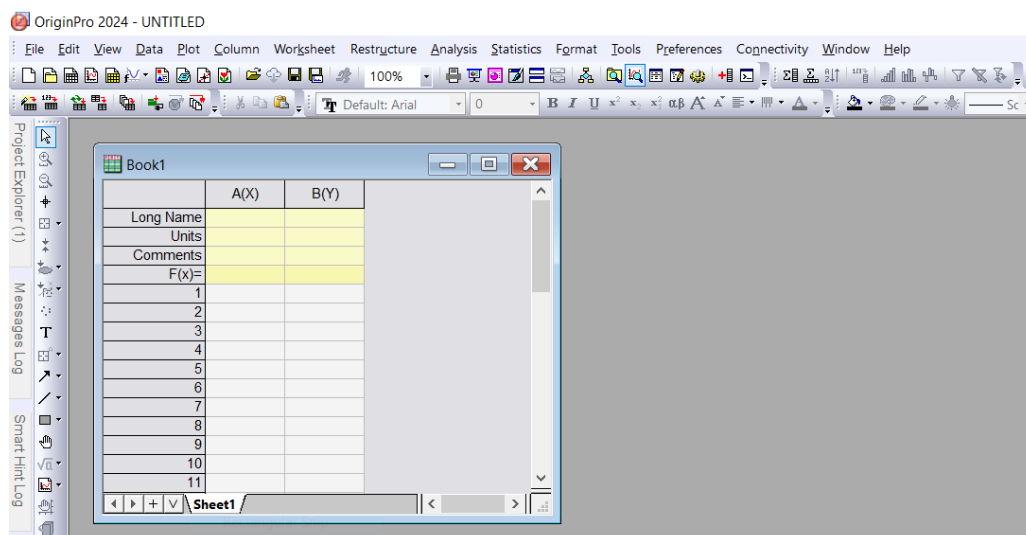
This would open up the Sample folder in a Windows Explorer.



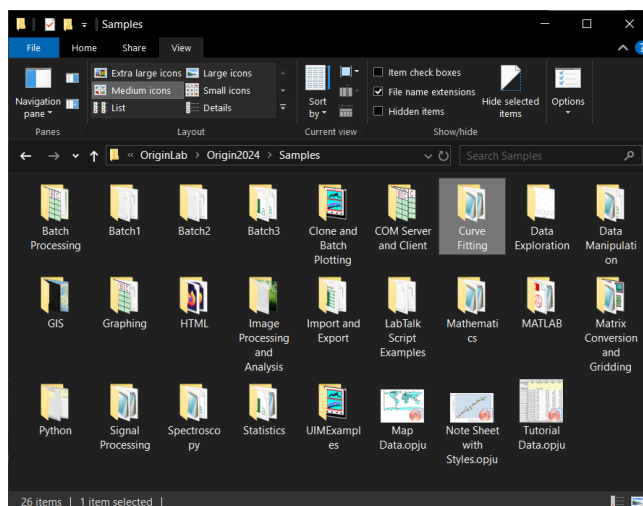
## 1.1 Plotting the first graph

### 1.1.1 Importing data from sample folder

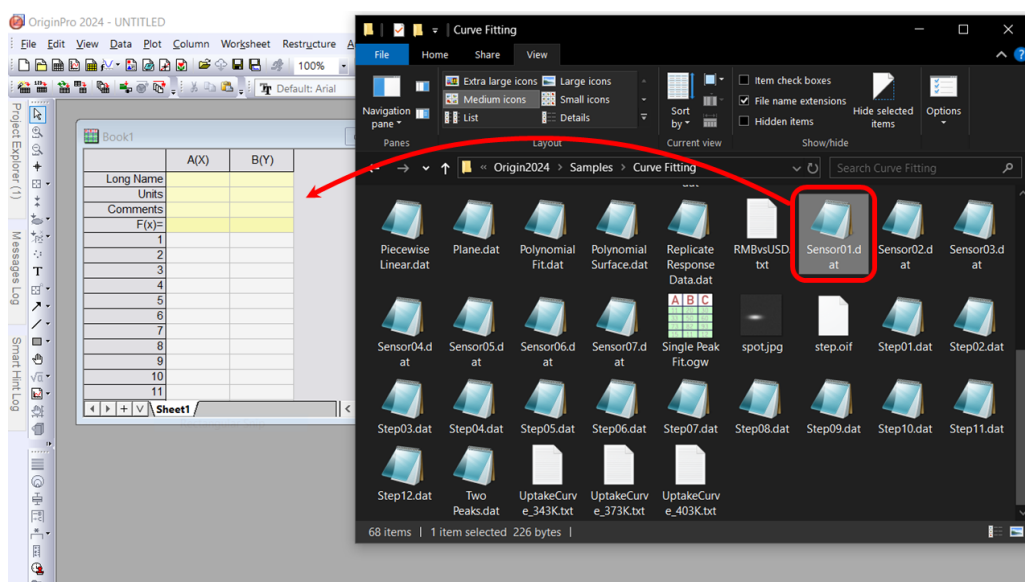
- I. Launch Origin to find the default empty workbook which contains a single worksheet with two columns.



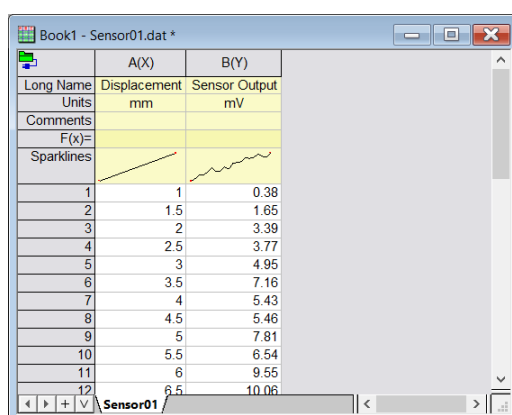
## II. Navigate into the Curve Fitting folder in the Sample folder.



## III. Locate any of the DAT files, such as Sensor01.dat. Then drag and drop it into the Origin workbook.

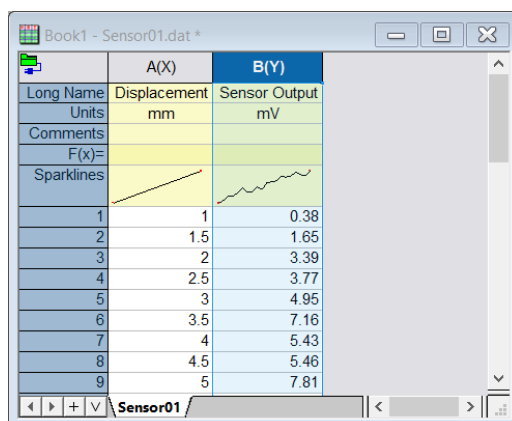


## IV. This would import the data from the DAT file into the workbook.



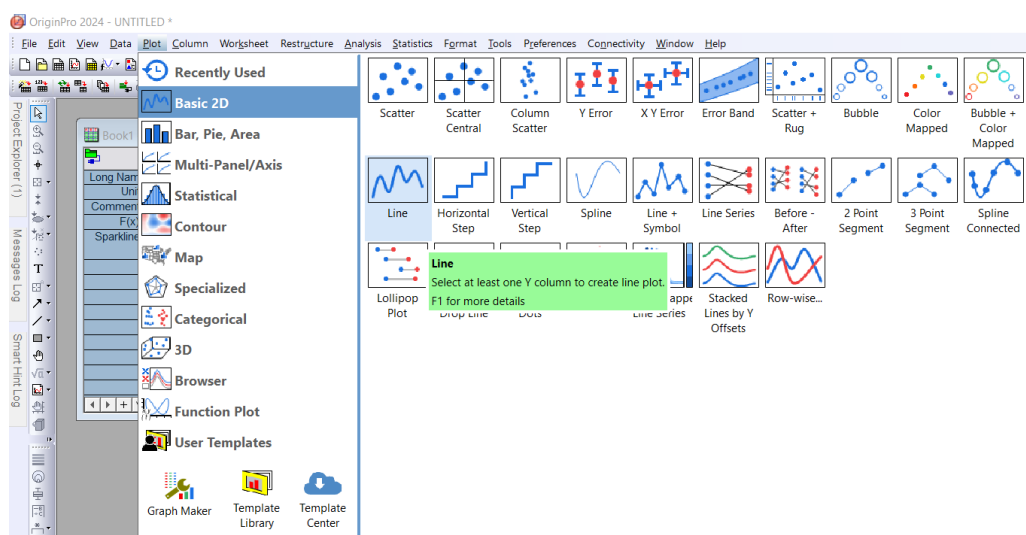
### 1.1.2 Plotting the graph

I. Click on the header of column B to select the entire column.

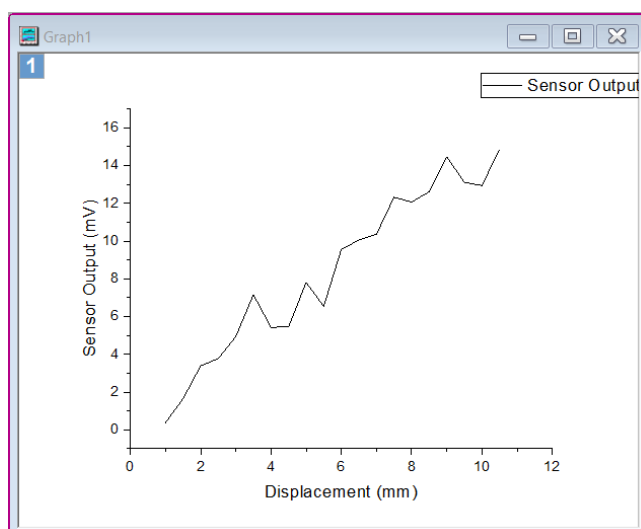


|            | A(X)         | B(Y)          |
|------------|--------------|---------------|
| Long Name  | Displacement | Sensor Output |
| Units      | mm           | mV            |
| Comments   |              |               |
| F(x)=      |              |               |
| Sparklines |              |               |
| 1          | 1            | 0.38          |
| 2          | 1.5          | 1.65          |
| 3          | 2            | 3.39          |
| 4          | 2.5          | 3.77          |
| 5          | 3            | 4.95          |
| 6          | 3.5          | 7.16          |
| 7          | 4            | 5.43          |
| 8          | 4.5          | 5.46          |
| 9          | 5            | 7.81          |

II. From the main menu, we click Plot > Basic 2D > Line.

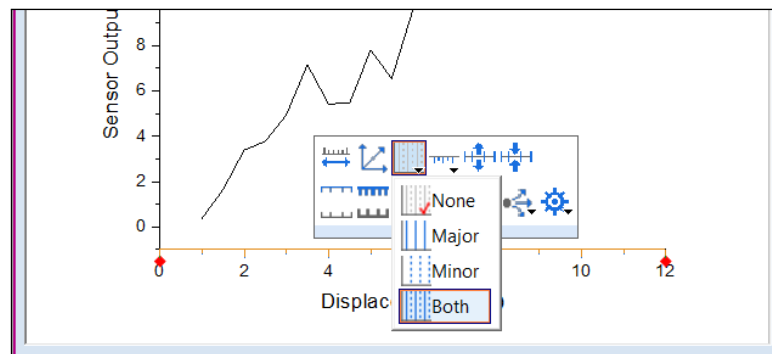


III. This produces a basic 2D plot.

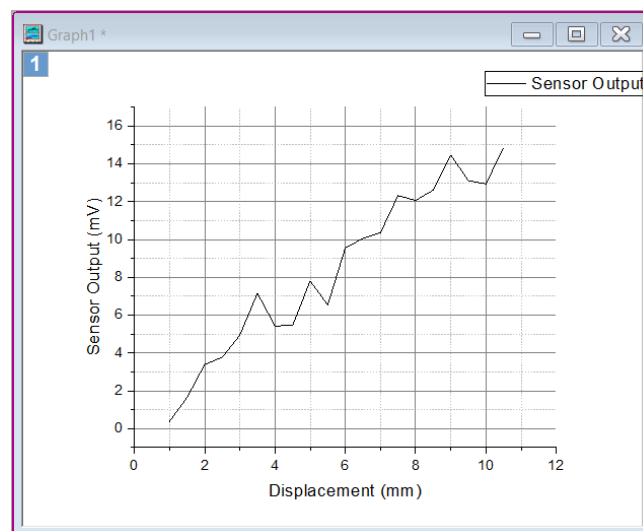


### 1.1.3 Basic customization of the plot

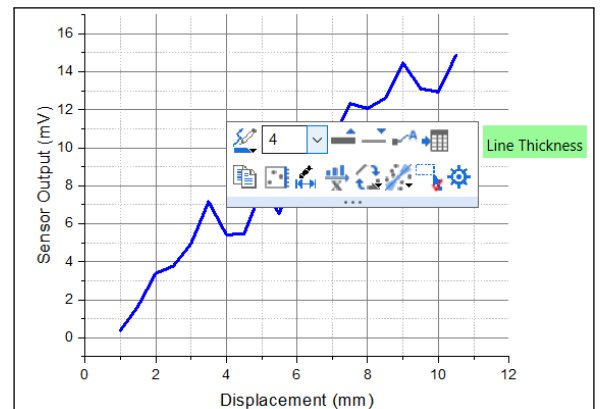
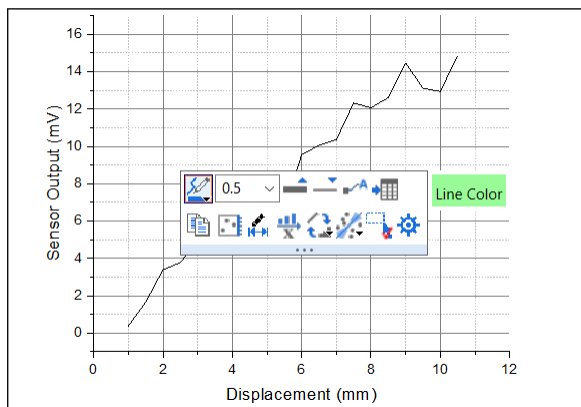
I. Click on the  $x$ -axis so that the mini toolbar pops up. Then click Show Gridlines > Both.



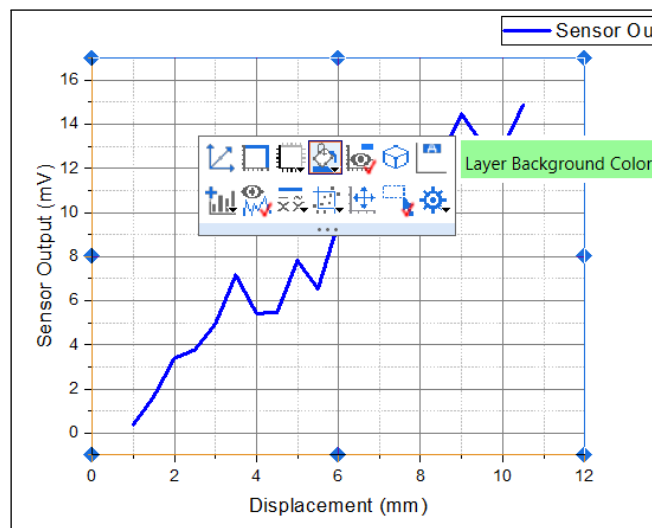
II. Repeat the above step on the  $y$ -axis to get a plot with major and minor gridlines.



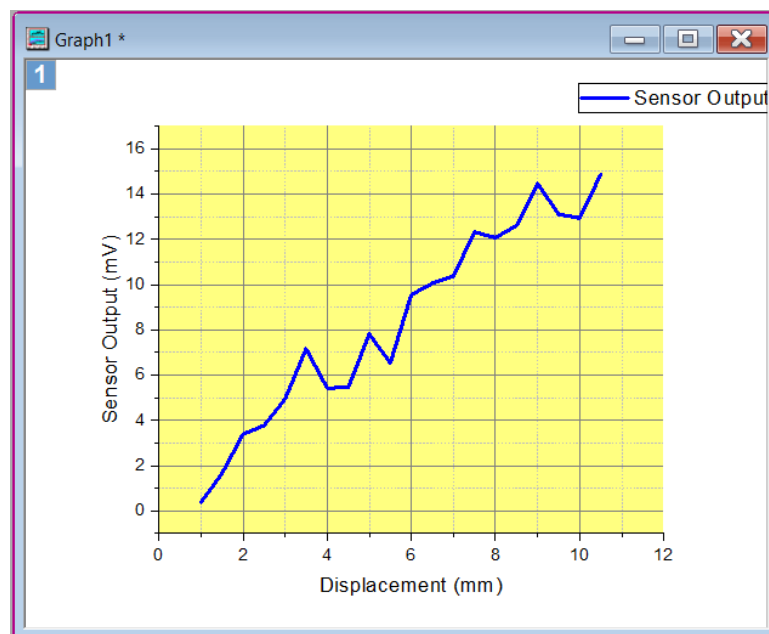
III. Click on the line plot so that the mini toolbar pops up. Then click on the Line Color button to select a colour, say blue. Also set the Line Thickness to a larger value, say 4.



- IV. Click on the plot background (ensure that the plot or the gridlines are not clicked), click on Layer Background Color and then select some background colour, such as light yellow.



- V. Finally we obtain a plot with better aesthetics.

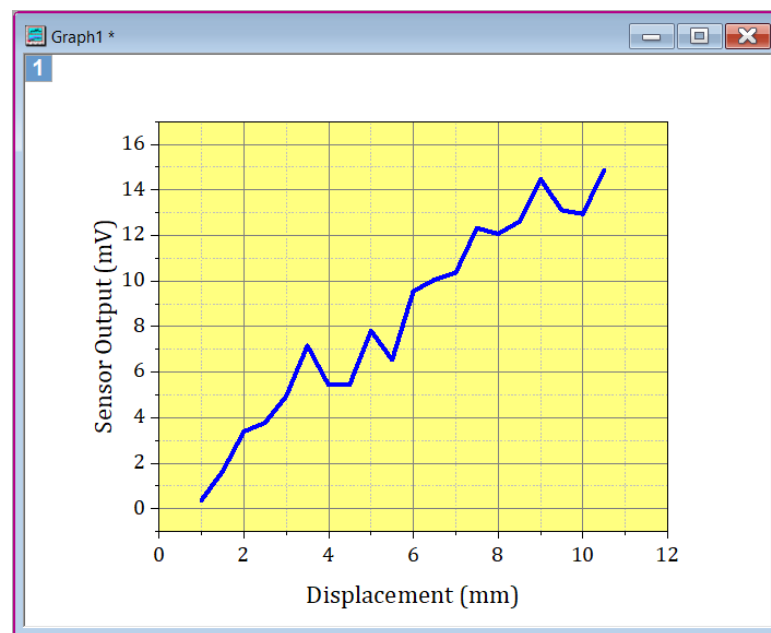


## 1.2 Using graph templates

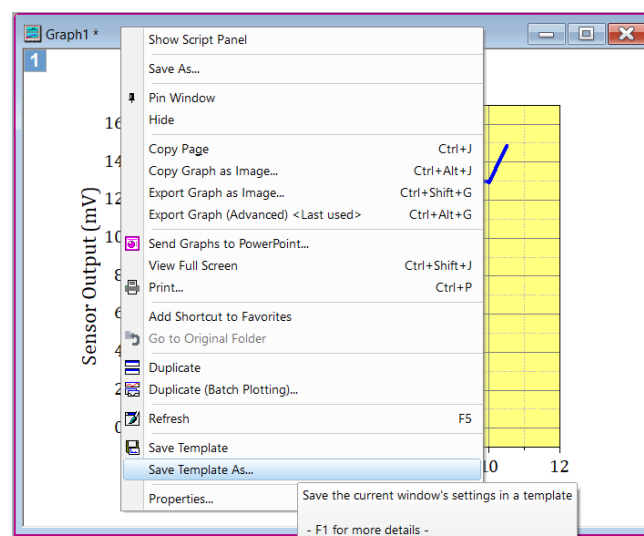
A template is a pre-made pattern, model, or mold used as a guide to make multiple identical or similar things. Origin allows us to prepare a graph template based on a graph which is customized according to our requirements. Then later, we can use this template to quickly make identical customizations on other graphs.

### 1.2.1 Creating a graph template

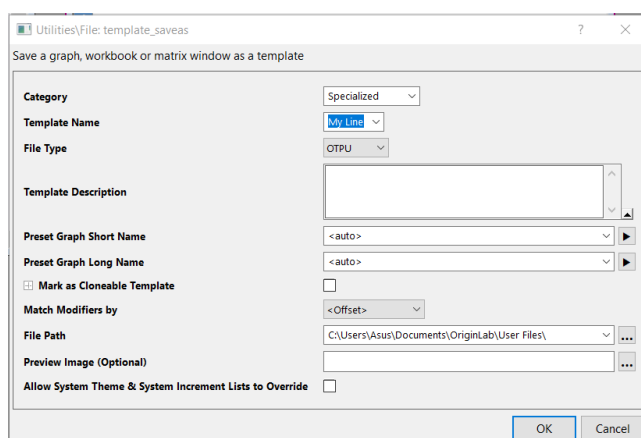
I. We start with a graph that we have customized as per our requirement.



II. To use this graph as a template, we right-click on the graph's title bar, and then click on Save Template As... from the context menu.

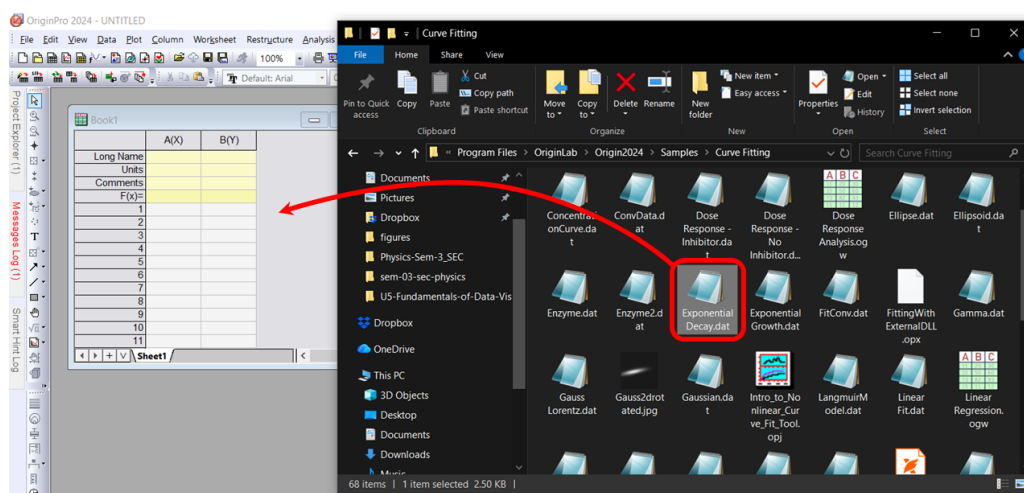


- III. In the save template dialog, we enter the name of the template, for example Template Name = My Line. Then we click OK.

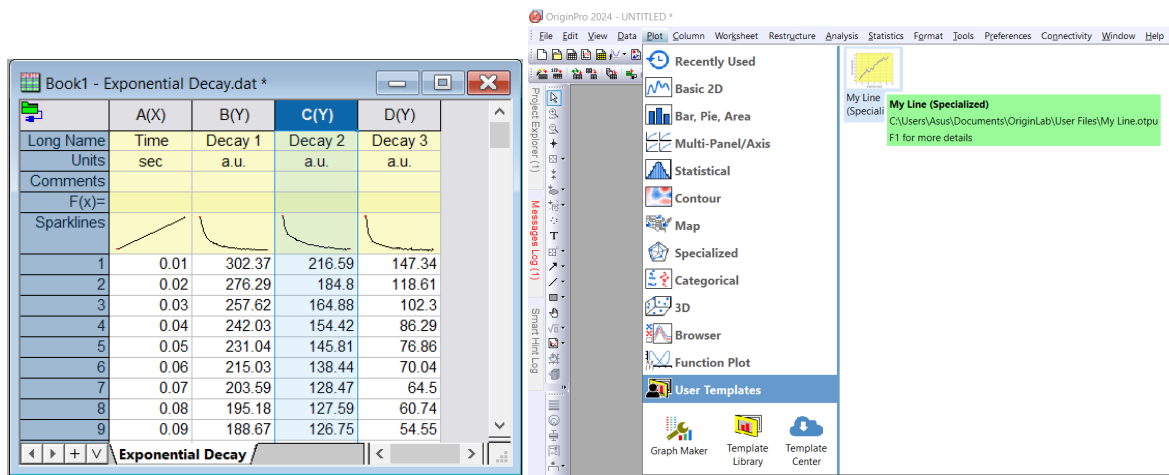


## 1.2.2 Applying a graph template

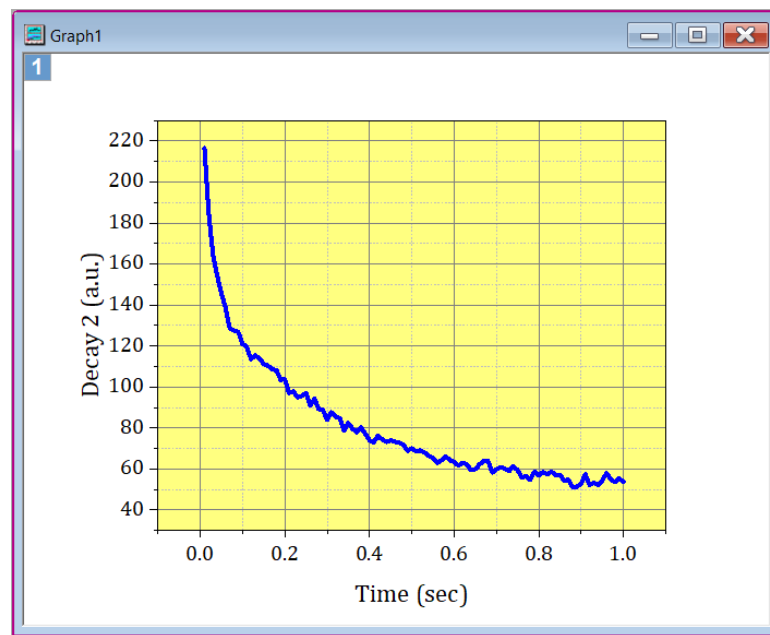
- Suppose in a different Origin workbook, we wish to plot some other graph using the above template. For example, we can drag and drop the Exponential Decay .dat file from the Samples > Curve Fitting folder into the workbook.



2. In order to plot the data in column C, we click on that column's header. Then from the main menu, we click Plot > User Templates > My Line.



3. This gives us the desired plot which is customized as per the saved graph template. We can find that the plotting style of this graph is identical to the one which we started with.



### 1.3 Batch plotting

## 2 Categorical and continuous data plots

A dataset is said to be *categorical* if its independent variable takes one of a small, discrete set of labels (e.g. the day of the week, the floor number of a skyscraper), and *continuous* if its independent variable can take any value within a range (e.g. time, angle, or magnetic field strength). The choice of plot type should follow from which of these two kinds of variable is being plotted, not from visual preference alone.

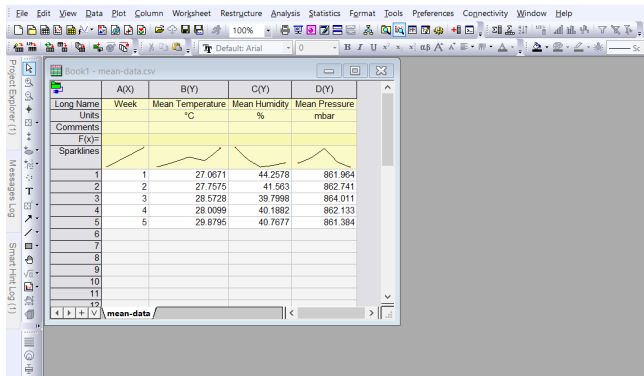
### 2.1 Column plots

In a *column plot*, each category is represented as a vertical bar whose height is proportional to a summary value (such as a mean, a total or a count).

#### 2.1.1 Creating a column plot in Origin

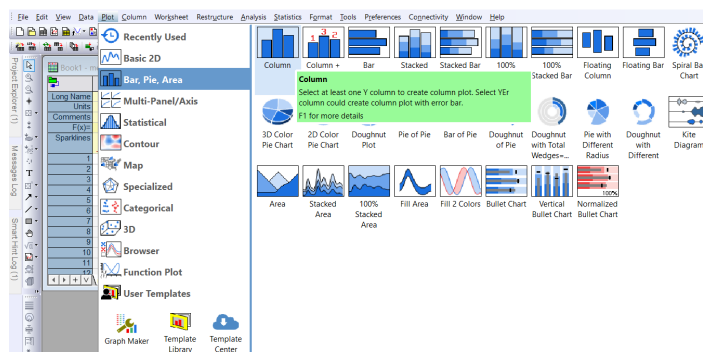
As a demonstration, we will create a column plot using the file named `mean-data.csv`, which has recorded the average temperature, humidity and pressure data for five weeks.

- I. We open the CSV file as described in the previous unit. To create the column plot for the mean temperatures for five weeks, first we select the entire column by clicking on the column name.

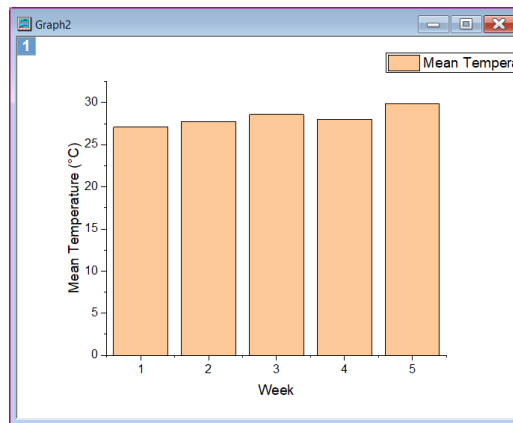


|            | A(X) | B(Y)             | C(Y)          | D(Y)          |
|------------|------|------------------|---------------|---------------|
| Long Name  | Week | Mean Temperature | Mean Humidity | Mean Pressure |
| Units      |      | °C               | %             | mbar          |
| Comments   |      |                  |               |               |
| F(x)=      |      |                  |               |               |
| Sparklines |      |                  |               |               |
| 1          | 1    | 27.0671          | 44.2578       | 861.964       |
| 2          | 2    | 27.7575          | 41.563        | 862.741       |
| 3          | 3    | 28.5728          | 39.7998       | 864.011       |
| 4          | 4    | 28.0099          | 40.1882       | 862.133       |
| 5          | 5    | 29.8795          | 40.7677       | 861.384       |
| 6          |      |                  |               |               |
| 7          |      |                  |               |               |
| 8          |      |                  |               |               |
| 9          |      |                  |               |               |
| 10         |      |                  |               |               |
| 11         |      |                  |               |               |
| 12         |      |                  |               |               |

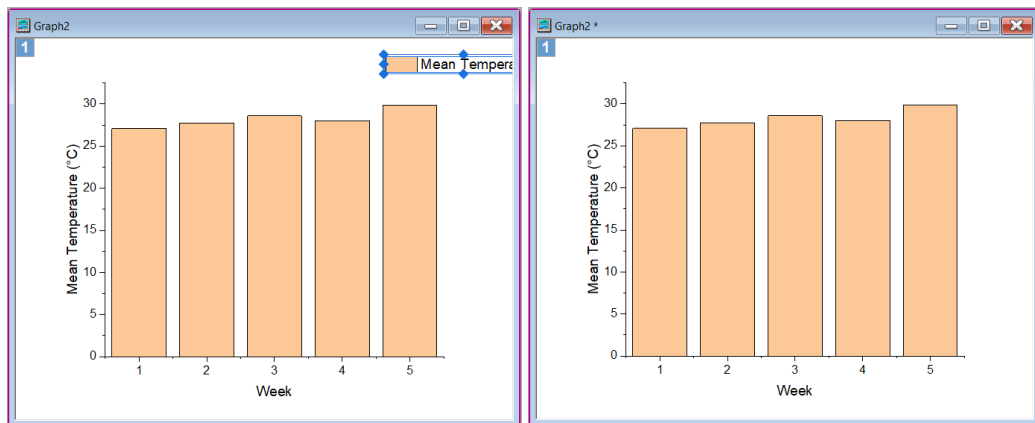
- II. From the main menu, we click Plot > Bar, Pie, Area, and then click Column.



III. This produces a rudimentary column plot with default settings.



IV. Because this is a column plot of only the mean temperature, we can click on the legend at the top right and press the delete button to remove it.



V. Because the column bars are about similar heights, we are unable to notice significant differences between them if we plot them from zero at the bottom. So we need to change the scale of the y-axis values in a manner such that their differences become more pronounced.

To do so, we click on any of the y-tick labels and then click the `Open Properties Dialog` logo.



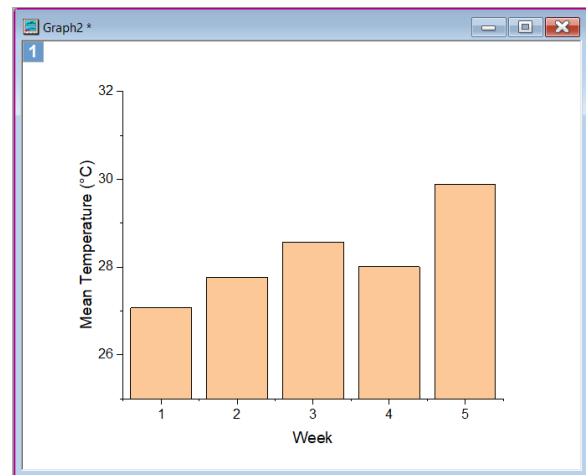
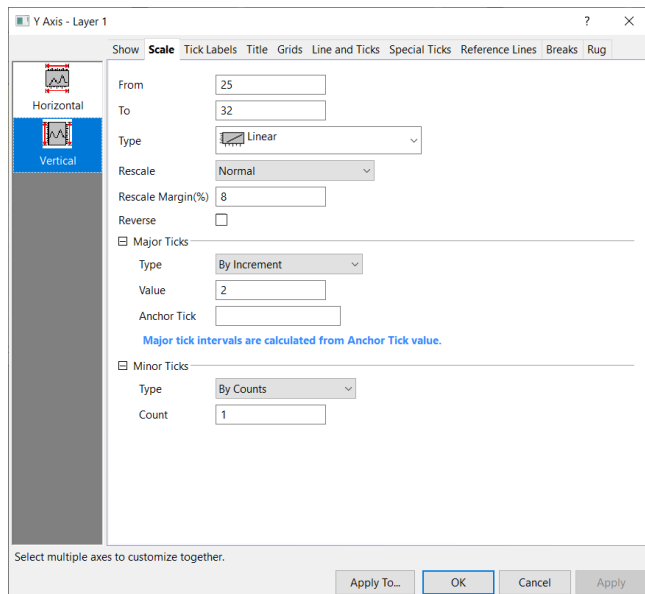
VI. In the Properties dialog, we click on the Scale tab and approximately set the following values as From = 25, To = 32, so as to ensure that the  $y$ -axis spans the values from 25 to 32.

We also set the major ticks as

Type = By Increment  
Value = 2

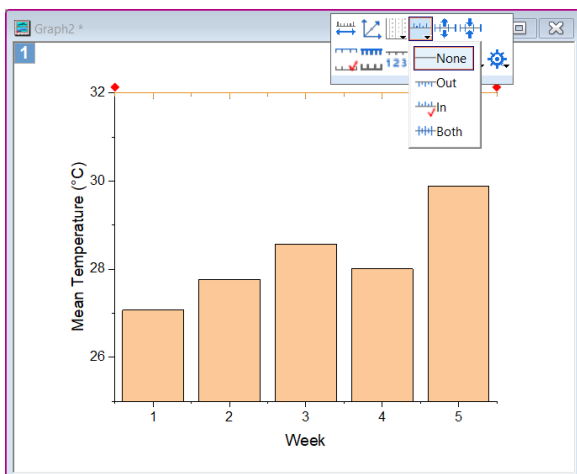
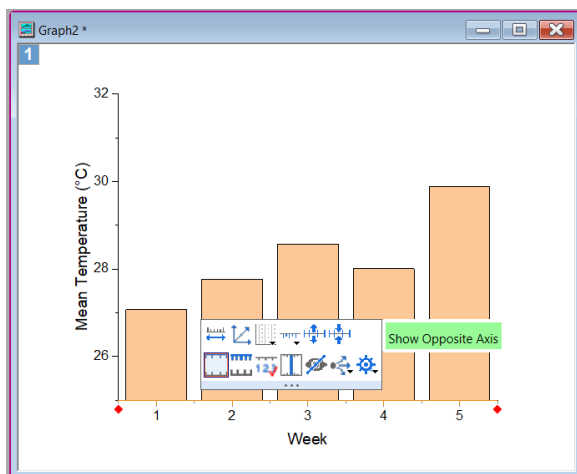
and the minor ticks as

Type = By Counts  
Value = 1

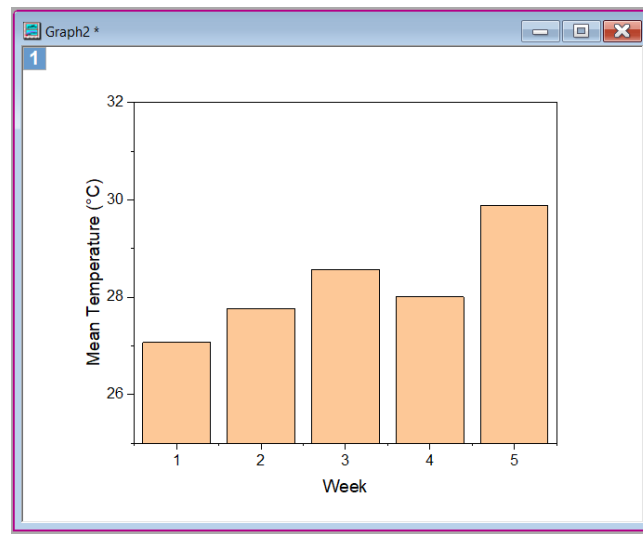


This shows much more discernible differences between the mean temperatures.

VII. Finally to place the graph within a rectangular bounding box, first we need to enable opposite axes by clicking on any of the axes lines, say the  $x$ -axis line, and then clicking on the Show Opposite Axis button, so that an identical axis appears at the top. Then we click on the new axis at the top, then click on Axis Type button and lastly click on None.



VIII. We repeat the same process for the  $y$ -axis as well to finally obtain the required column plot within a bounding box.



## 2.2 Line plots

A *line plot* connects successive data points with straight (or smoothed) line segments. It is the standard choice for a continuous independent variable, most commonly time. It should not be used for categorical data, since a line drawn between two unrelated categories implies a continuity between them that does not exist.

## 2.3 Pie charts

A *pie chart* represents each category's share of a whole as a wedge of a circle, with wedge angle proportional to that category's proportion of the total. Pie charts are appropriate only for categorical data that sums to a meaningful whole, and are best limited to a small number of categories (three to five); beyond this, a column chart is usually clearer.

## 2.4 Area plots

An *area plot* is a line plot with the region between the line and the axis filled in. It is used, in place of an ordinary line plot, when the magnitude of a continuously-varying quantity is the point of emphasis, or when several such quantities are to be compared by their relative contribution to a stacked total.

## 2.5 Scatter plots

A *scatter plot* represents each observation as an individual point, with no line joining successive points. It is the correct choice when the relationship between two continuous variables is of interest, rather than the trend of one variable against time. A scatter plot is particularly used when

there is no reason to assume the points are ordered or continuous in the underlying physical process.